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ERICPOL
i-EVOLUTION

ERICSSON

Rola kryptografii krzywych eliptycznych w bezpieczeństwie systemów teleinformatycznych

$$F_p = \{0, 1, 2, \ldots, p-1\}$$

$$F_7 = \{0,1,2,3,4,5,6\}$$

$$a = b \pmod{p}$$

$$a - b = q * p$$

$$a = b \pmod{p}$$

10=3 (mod 7)

10-3=7

$$a = b \pmod{p}$$

10=3 (mod 7)

10-3=7

19=5 (mod 7)

19-5=14

$$a = b \pmod{p}$$

10=3 (mod 7)

10-3=7

19=5 (mod 7)

19-5=14

23=4 (mod 7)

23-4=19 X

$$a+b\left(mod\ p\right)=r$$

$$a+b=p*q+r$$

$$a+b=p*q+r$$

p=7

8+6=14 + 0

8+6 (mod 7)=0

$$a+b=p*q+r$$

$$8+6=14+0$$

$$8+6 \pmod 7=0$$

$$10+3=7+5$$

$$10+3 \pmod 7=5$$

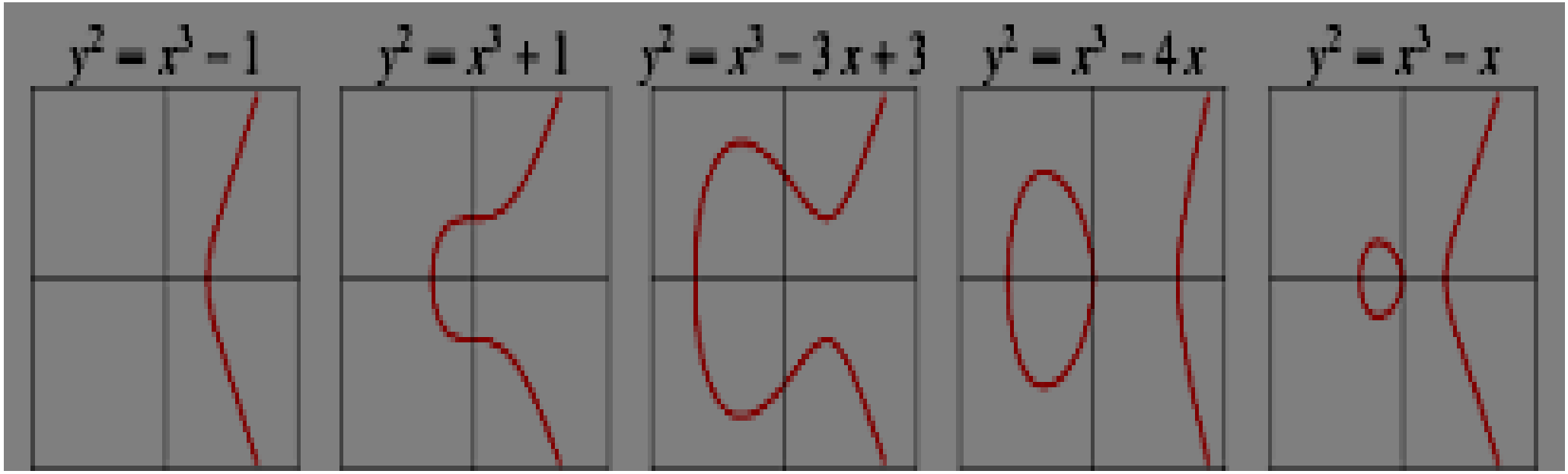
$$r \in F_p$$

$$F_p = \{0, 1, 2, \ldots, p-1\}$$

$$a * b \pmod{p} = r$$

$$y^2 = x^3 + ax + b$$

- Przykłady krzywych



$$F_7 = \{0,1,2,3,4,5,6\}$$

(0,0)(0,1)(0,2)(0,3)...(0,6)

(1,0)(1,1)(1,2)(1,3)...(1,6)

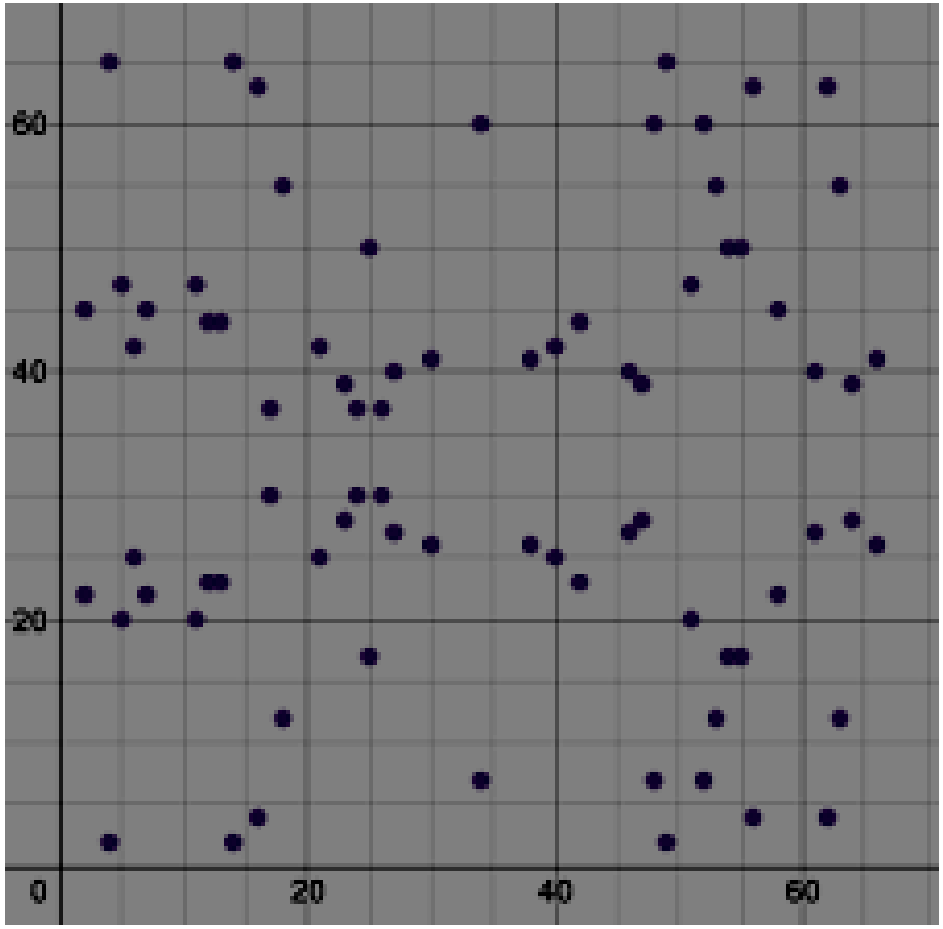
...

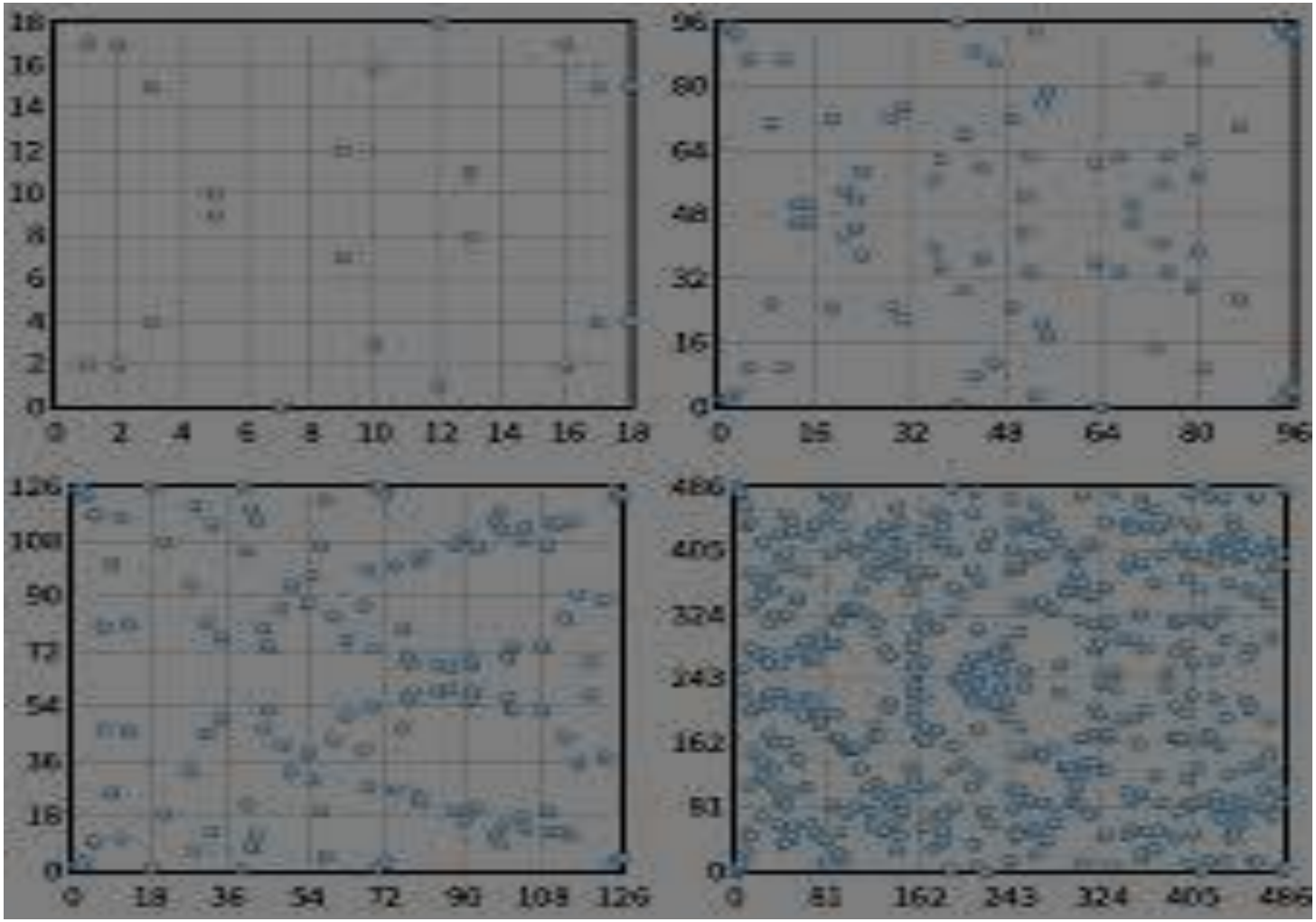
(6,0)... (6,6)

$$y^2 = x^3 + ax + b$$

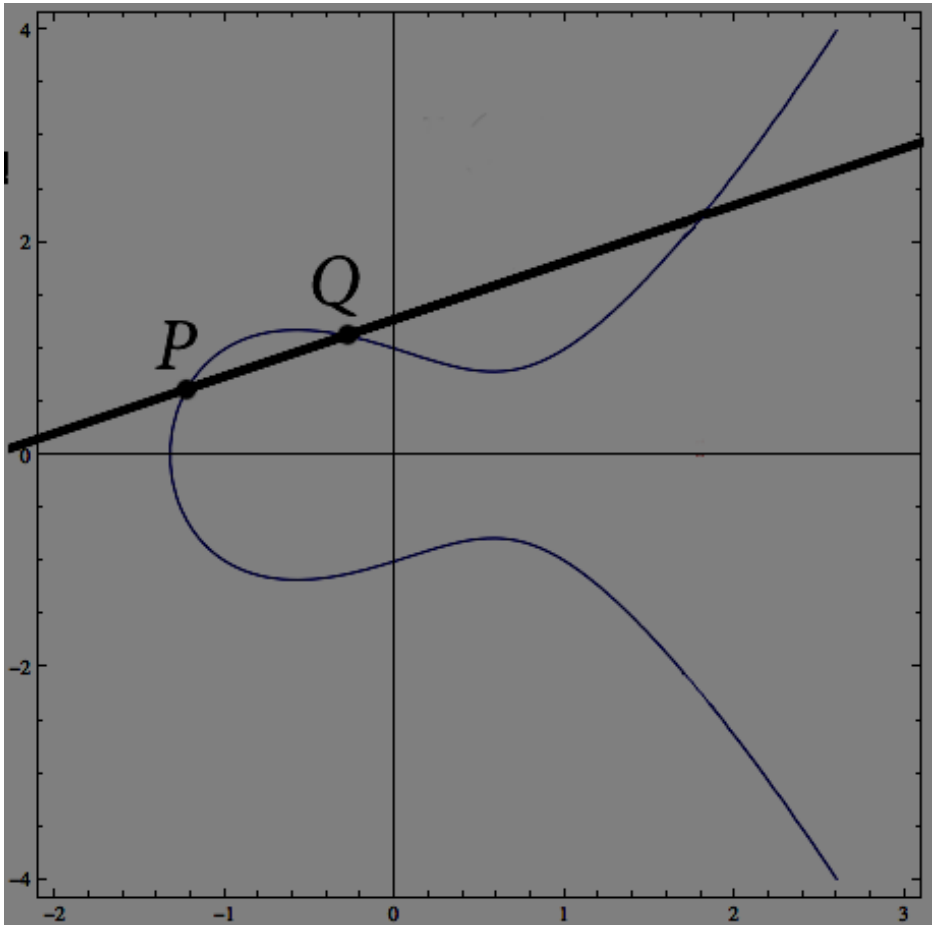
(1,2) x=1, y=2

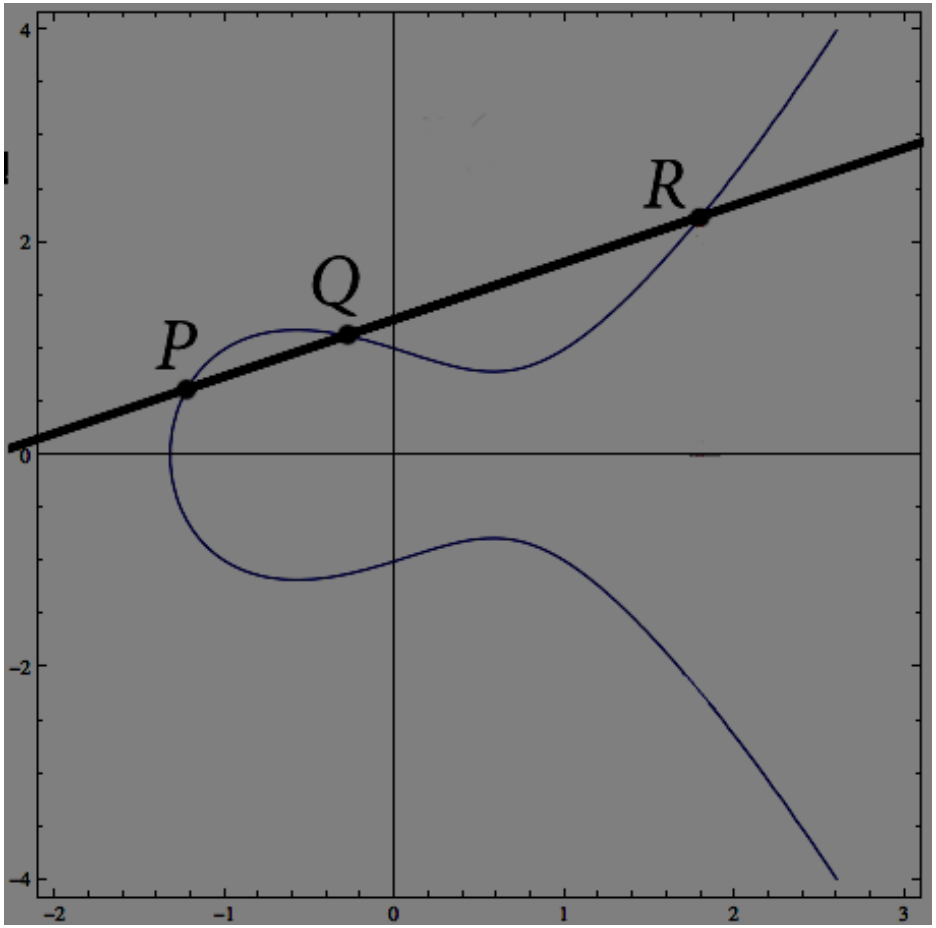
E(Fp)

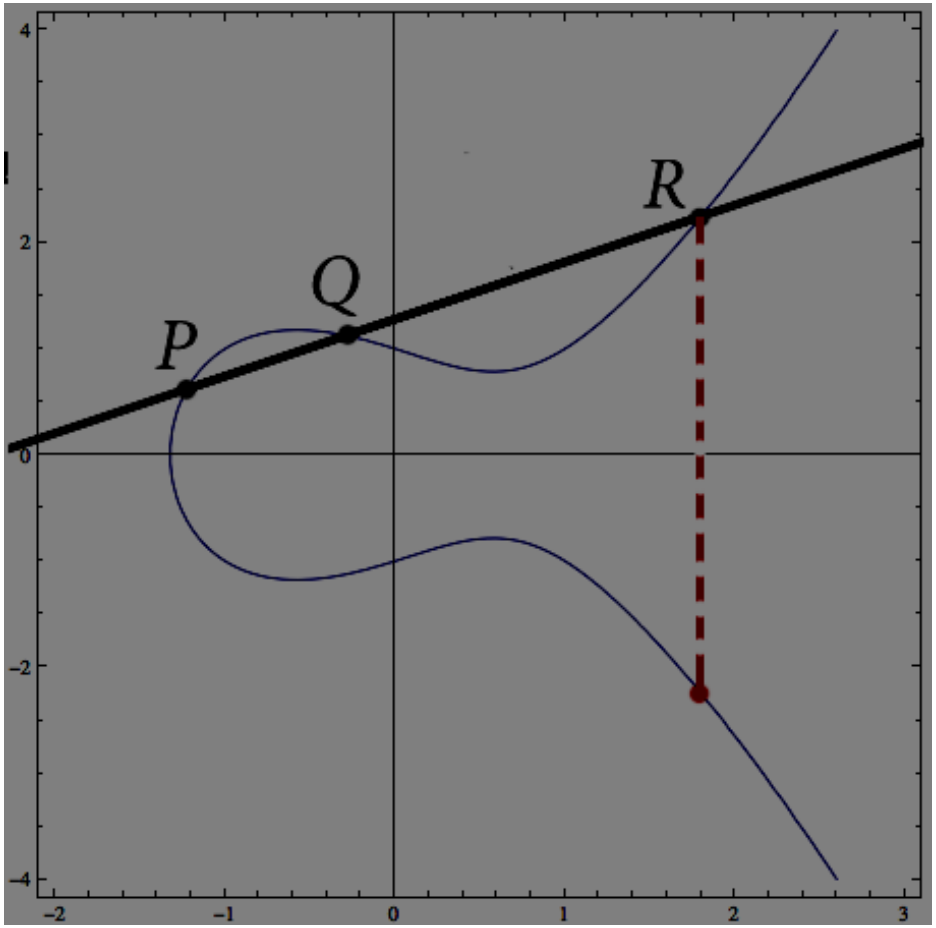


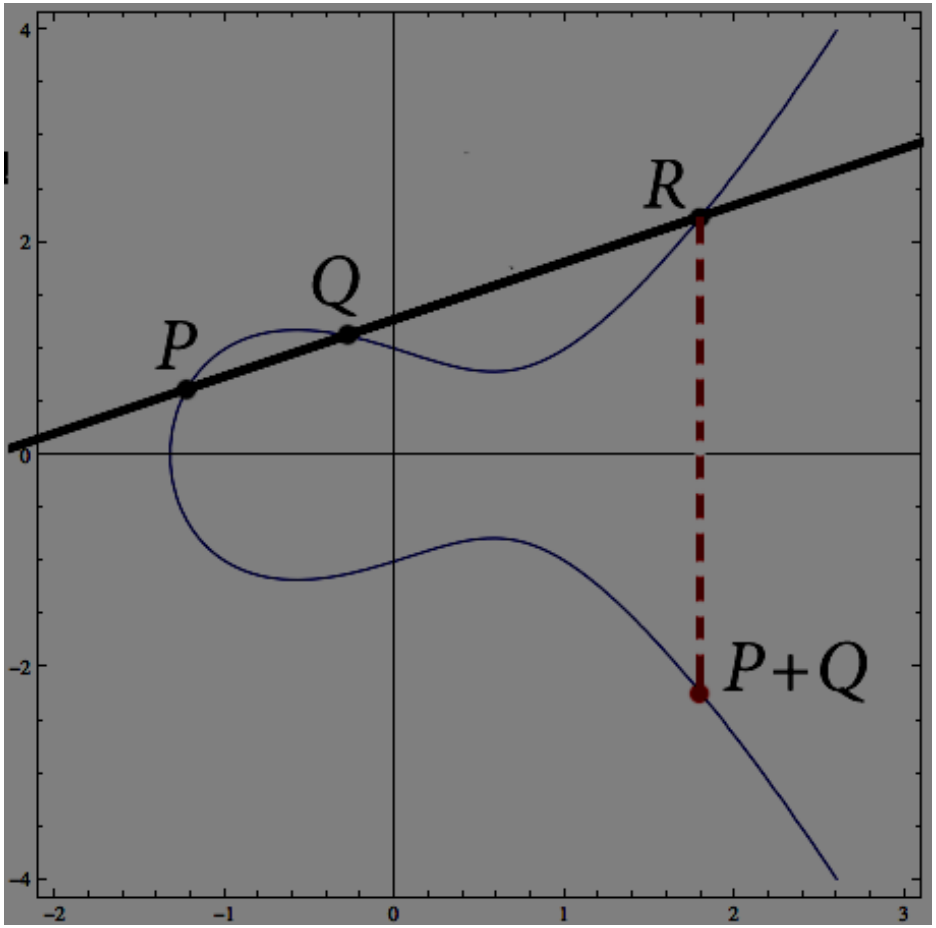


$$y^2 = x^3 + ax + b$$ & Fp


P
Q
4
2
0
−2
−4
−2
−1
0
1
2
3

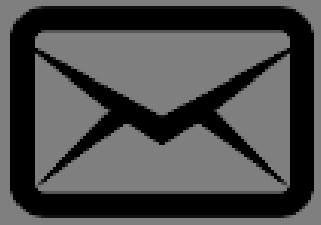

P
Q
R
4
2
0
−2
−4
−2
−1
0
1
2
3


R
Q
P
4
2
0
−2
−4
−2
−1
0
1
2
3


R
Q
P
P+Q
4
2
0
−2
−4
−2
−1
0
1
2
3

El Gamal

E, Fp



$\rightarrow P_m$

El Gamal

E, Fp

$\rightarrow P_m$

a

B, aB

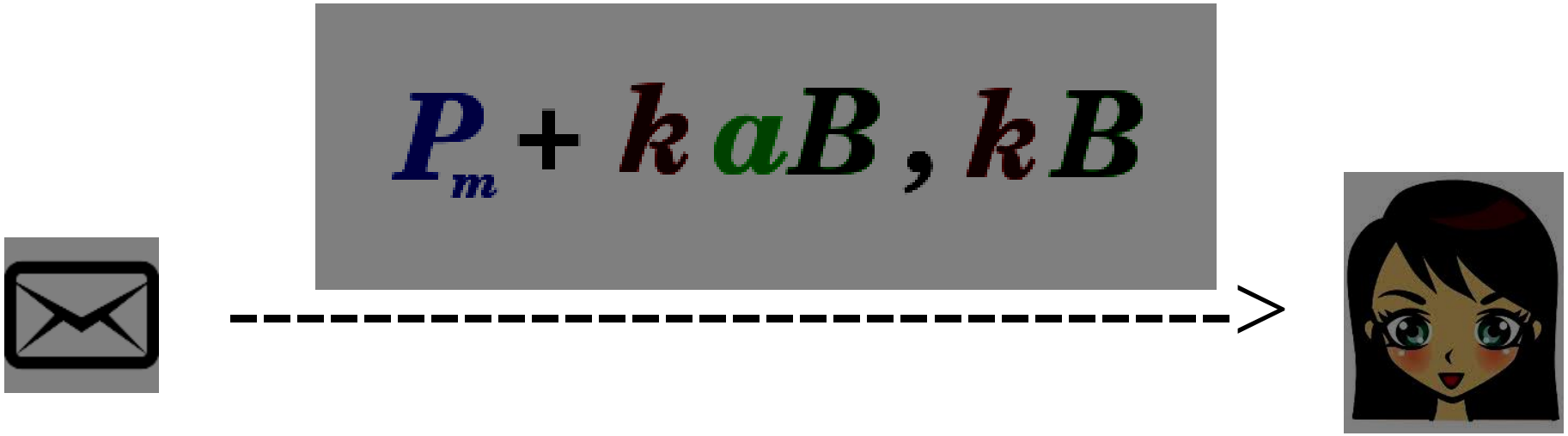
El Gamal - szyfrowanie

$$k$$

$$kaB$$

$$kB$$

$$P_m + kaB$$


$P_m + kaB, kB$

El Gamal - rozszyfrowanie

$$P_m + kaB, kB$$

$$akB$$

$$P_m + kaB - akB$$

$$kB=B+B+B+\ldots + Bk$$

El Gamal → El Gamal EC

DH → ECDH

DSA → ECDSA

What 5G is about

European Commission

Smart card

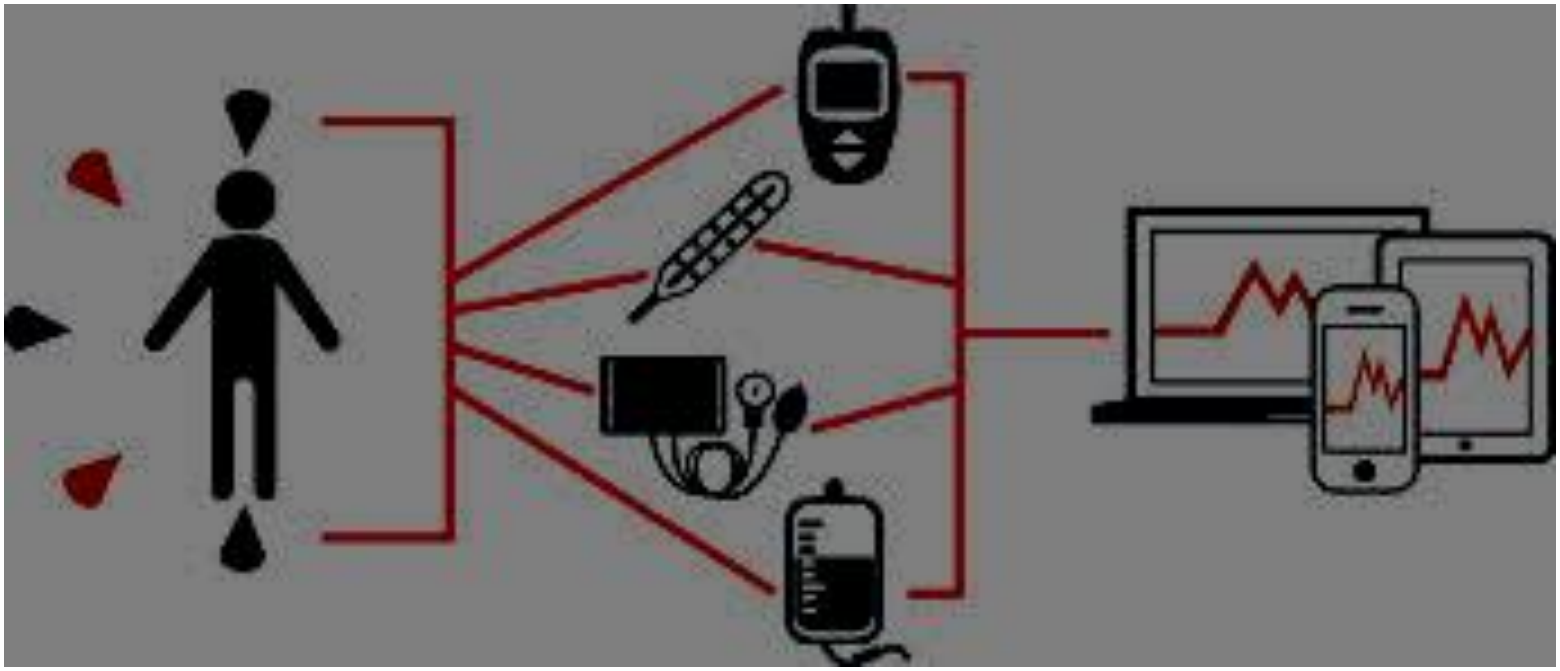
THE BANK
JOHN SMITH
JOHN SMITH
JOHN SMITH
JOHN SMITH

THE UNIVERSITY OF HONG KONG
Staff Card

Smart home

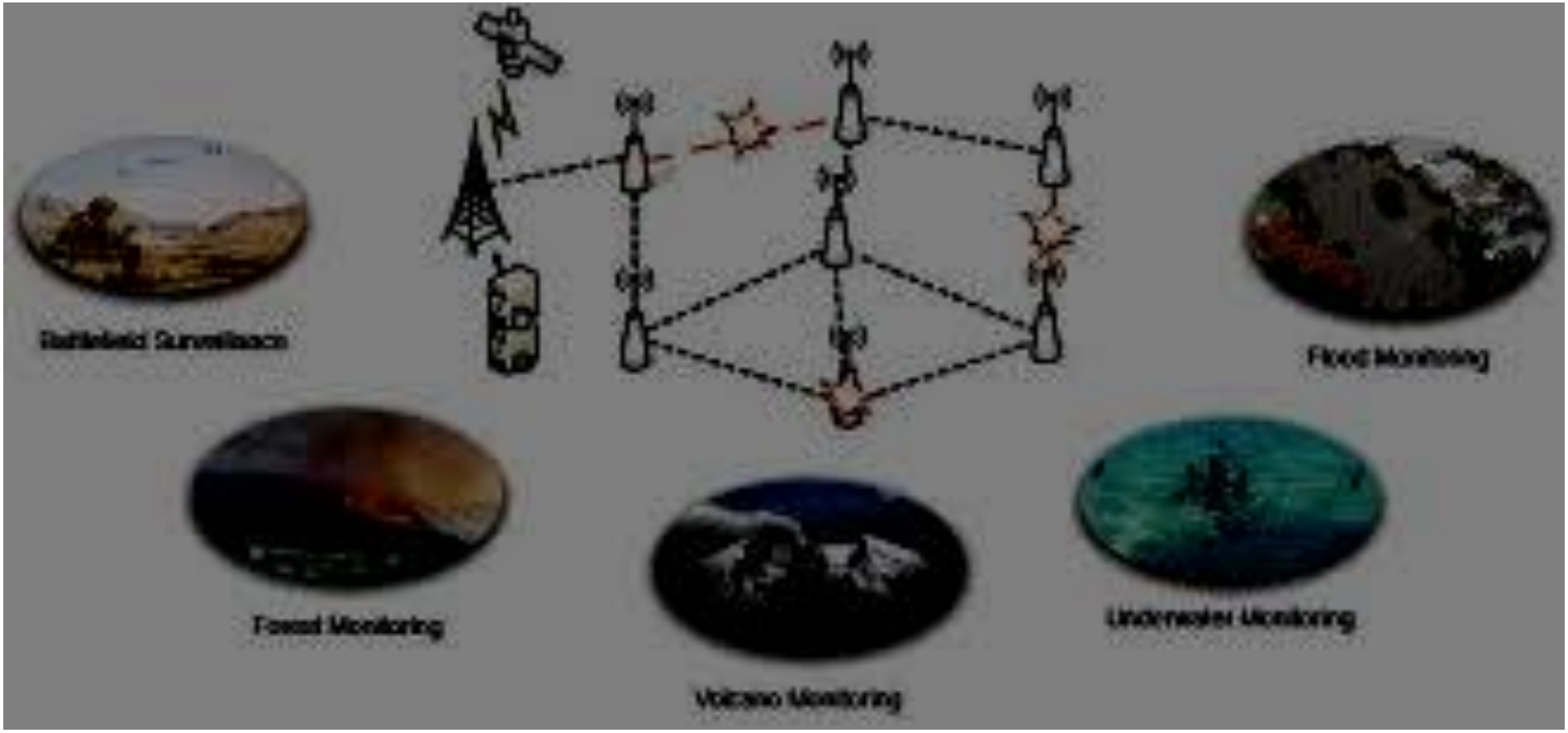
AUTO CURTAIN
HOME ENTERTAINMENT
COMMUNICATION
WOODS
HOME AUTOMATION
GARAGE
AUTO SPRINKLERS
ON
OFF
AUTO LIGHTING
HOME BACK UP SYSTEM
SECURITY
WCAM

eHealth



Smart transport

Sensor networks



Symmetric Key Size (bits)	RSA and Diffie-Hellman Key Size (bits)	Elliptic Curve Key Size (bits)
80	1024	160
112	2048	224
128	3072	256
192	7680	384
256	15360	521

Table 1: NIST Recommended Key Sizes

Rekomendacje

Dziękuję

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